Customer Shopping Behavior Analysis

An End-to-End Data Analytics Project

by: Saroj kumar Nayak

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Project Overview: Problem & Objective

Problem Statement

Retailers struggle to understand customer behavior, product preferences, and sales drivers. This leads to inefficient marketing, suboptimal inventory, and lost revenue.

Project Objective

Analyze shopping data to understand demographics, identify top categories, analyze purchasing patterns, and generate actionable insights.

Analytics Lifecycle: Python → EDA → SQL → Dashboard → Recommendations

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Dataset Description & Summary

Key Columns

Customer ID, Age, Gender

Demographic & unique identifiers

Category, Purchase Amount

Product & transaction details

Subscription, Review, Frequency

Behavioral & satisfaction metrics

Data Snapshot

- ~3,900+ Records
- 10+ Columns
- Mixed Data Types: Numerical & Categorical
- Main Categories: Clothing, Accessories, Footwear, Outerwear

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Data Pre-Processing: Ensuring Quality

Data Cleaning

- Missing values check
- Data type verification
- Consistency in labels
- Duplicate removal

Data Engineering

- Age group creation
- Categorical standardization
- SQL data structure prep
- Column naming consistency

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Data Analysis Summary (EDA + SQL Insights)

Exploratory Data Analysis – Key Insights

- Customer base is **balanced across age groups**, indicating broad market coverage.
- **Clothing** is the most purchased category across all genders and age segments.
- **Male customers** contribute a higher number of transactions compared to females.
- Most customers purchase **weekly to monthly**, showing strong repeat behavior.
- A large portion of users are **subscribed customers**, supporting retention-driven revenue.

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SQL Data Analysis – Key Business Insights

- **Gender Revenue Analysis:**
  Revenue contribution differs by gender, helping identify high-value customer segments.
- **Discount Effectiveness:**
  Several customers spend **above average even with discounts**, proving discounts can drive higher-value purchases.
- **Top-Rated Products:**
  Products with higher average review ratings indicate **strong customer satisfaction** and quality.
- **Shipping Impact:**
  Purchase amounts vary between **Standard and Express shipping**, reflecting delivery preference influence.
- **Subscription Value:**
  Subscribed customers generate **higher average spend and total revenue**, validating the subscription model.
- **Discounted Product Trends:**
  Certain products consistently receive higher discounts, indicating **promotion-driven demand**.
- **Customer Segmentation:**
  Customers are effectively segmented into **New, Returning, and Loyal**, enabling targeted strategies.
- **Category Leaders:**
  Top products differ by category, helping identify **best-performing items** in each segment.
- **Repeat Buyers & Subscriptions:**
  Repeat buyers show a higher likelihood of being subscribed, confirming loyalty behavior.
- **Age Group Revenue Contribution:**
  Revenue varies by age group, supporting **age-based marketing and personalization**.

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Dashboard & Data Visualization

Tools Used

- Dashboard: Power BI
- Design: Figma (custom background)

Dashboard

- KPIs: Total Purchase, Avg. Review, Total Customers
- Interactive Filters: Gender, Category, Subscription
- Visuals: Donut, Bar, Tree maps, Pie charts

Professional UI/UX for visual consistency and storytelling.

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Key Business Insights & Recommendations

Insights

- Clothing: Top-performing category
- Male customers: Higher purchase volume
- Subscription users: Consistent purchases
- Diversified customer base
- High ratings: Good satisfaction

Recommendations

- Increase subscription conversion
- Focus on high-revenue categories (Clothing, Accessories)
- Improve low-performing categories (Outerwear)
- Customer segmentation for targeted campaigns
- Retention strategies for loyal customers

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Conclusion: End-to-End Analytics

Data Prep

Cleaning & EDA in Python

SQL Analysis

Analytical querying

Insights

Business-driven recommendations

Visualization

Power BI & Figma

Demonstrates real-world analytics skills for portfolios, internships, and job interviews.

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Thank you

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